

WEEK-4 BA TASK

High-Performance Market Basket Analytics with Compressed FP-Trees from Scratch

Task Overview

Intha task-oda main goal, transaction data-la customers frequently serthu vaangura products-a kandupidikkaradhu. Adhukkaga FP-Tree-ai from scratch build panni, node-link pointers, conditional FP-Trees, frequent patterns matrum high-confidence association rules generate panna vendum.

1. Objective

Market Basket Analytics use panni frequent item combinations-ai identify pannuvom. FP-Tree compressed representation use pannuradhunaala repeated transaction prefixes-ai share panni memory usage-ai reduce panna mudiyum. Large transactional datasets-kum efficient-ah work aagura mining engine create pannuvadhu task-oda objective.

2. Key Concepts

Concept	Simple Tanglish Meaning
FP-Tree	Transactions-ai compressed tree structure-la store pannum.
Header Table	Frequent items and avanga node links-ai maintain pannum.
Node Link	Same item-oda different tree nodes-ai connect pannum pointer.
Conditional FP-Tree	Oru particular suffix/item-ai base panni create pannura smaller FP-Tree.
Support	Oru itemset ethana transactions-la varudhu enbadhoda frequency.
Confidence	Rule-la left side nadandha, right side nadakkura probability.
Association Rule	Example: Bread -> Butter.

3. Workflow

Dataset -> Preprocessing -> Item Frequency Count -> FP-Tree Construction -> Header Table + Node Links -> Conditional Pattern Bases -> Conditional FP-Trees -> Frequent Itemsets -> Association Rules -> Confidence Evaluation -> Performance Analysis

4. Sample Transaction Dataset

TID	Items
T1	Milk, Bread, Butter
T2	Bread, Butter
T3	Milk, Bread
T4	Milk, Bread, Butter
T5	Bread, Butter, Jam
T6	Milk, Bread, Jam

5. Python Implementation

```
from collections import Counter
```

```
class FPNode:
    def __init__(self, item, count, parent):
        self.item = item
```

```

        self.count = count
        self.parent = parent
        self.children = {}
        self.link = None

def build_fp_tree(transactions, min_support=2):
    freq = Counter(item for t in transactions for item in t)
    freq = {i:c for i,c in freq.items() if c >= min_support}

    root = FPNode(None, 0, None)
    header = {i: [freq[i], None] for i in freq}

    for transaction in transactions:
        items = [i for i in transaction if i in freq]
        items.sort(key=lambda x: (-freq[x], x))

        current = root
        for item in items:
            if item not in current.children:
                node = FPNode(item, 0, current)
                current.children[item] = node

            if header[item][1] is None:
                header[item][1] = node
            else:
                n = header[item][1]
                while n.link is not None:
                    n = n.link
                n.link = node

            current.children[item].count += 1
            current = current.children[item]

    return root, header

transactions = [
    ["Milk", "Bread", "Butter"],
    ["Bread", "Butter"],
    ["Milk", "Bread"],
    ["Milk", "Bread", "Butter"],
    ["Bread", "Butter", "Jam"],
    ["Milk", "Bread", "Jam"]
]

root, header = build_fp_tree(transactions, min_support=2)

for item, (support, node) in header.items():
    print(item, "Support:", support)

```

6. Association Rule Calculation

Example rule: Bread -> Butter. Confidence formula: $\text{Confidence}(\text{Bread} \rightarrow \text{Butter}) = \frac{\text{Support}(\text{Bread, Butter})}{\text{Support}(\text{Bread})}$. Sample dataset-la Bread 6 transactions-la irukku; Bread + Butter 4 transactions-la irukku. So confidence = $4/6 = 66.67\%$.

7. Performance Optimization

- Repeated prefixes-ai FP-Tree compression moolama share pannuvom.
- Low-frequency items-ai early filtering panni tree size reduce pannuvom.
- Header table + node links use pannina same item nodes-ai quickly traverse panna mudiyum.
- Conditional trees original dataset-ai repeatedly scan panna vendiya requirement-ai reduce pannum.
- Large datasets-la execution time and memory usage measure panni compare pannalaam.

8. Expected Output

Program frequent items and their support values-ai display pannum. FP-Tree build aagum; header table-la node links maintain aagum. Adhukku mela frequent itemsets and high-confidence association rules generate panna mudiyum.

9. Conclusion

Intha project moolama FP-Tree based frequent pattern mining-ai from scratch understand panna mudiyum. Compressed tree, node-link pointers and conditional FP-Trees use pannuradhunaala market basket analysis-ai efficient-ah perform panna mudiyum. Association rules business-la product recommendation, cross-selling and inventory planning-ku useful-ah irukkum.